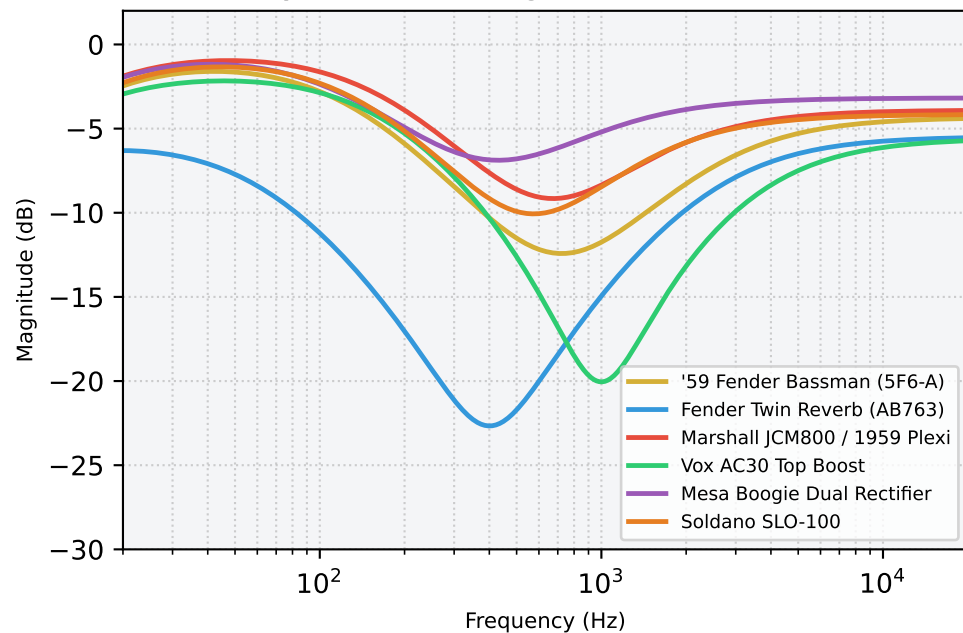


# CLASSIC AMP TONE STACK COMPARATIVE BENCHMARK

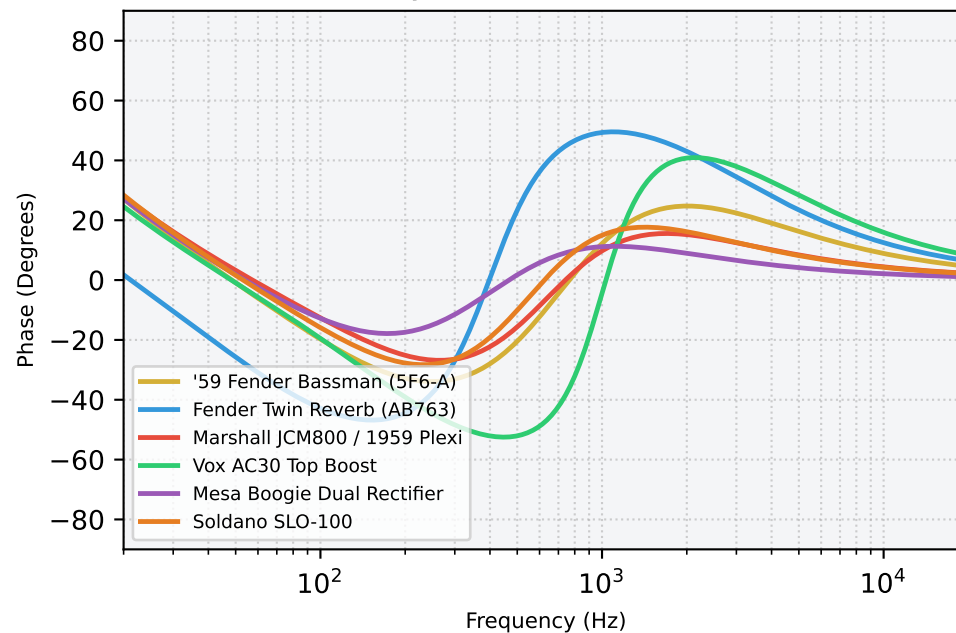
Mathematical Comparison of 6 Iconic Circuit Topologies (Fender, Marshall, Vox, Mesa, Soldano)

Yeh & Smith DAFx-06 Nodal Synthesis

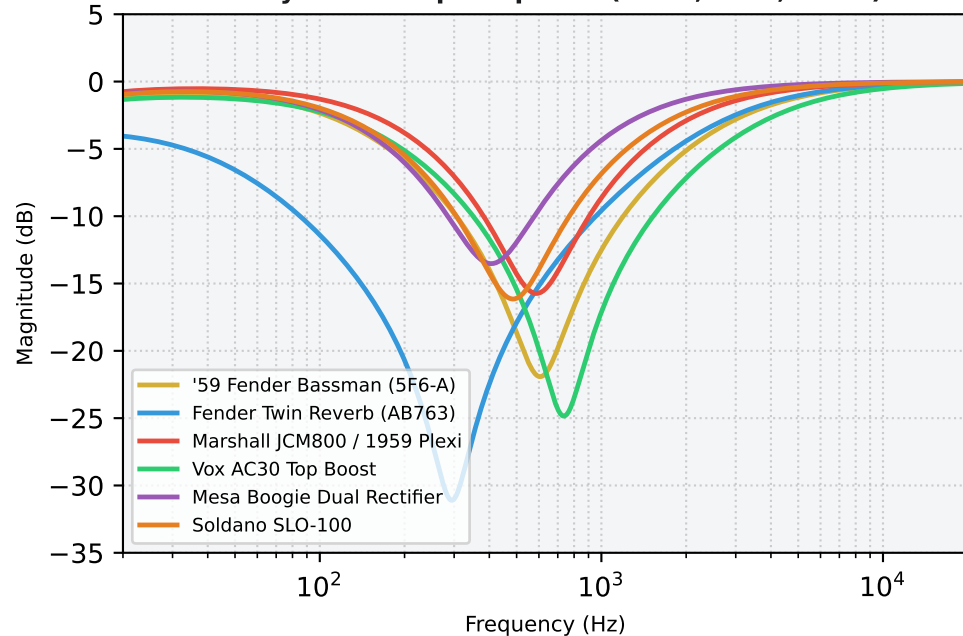
Factory Default Tone Signatures (T=5, M=5, B=5)



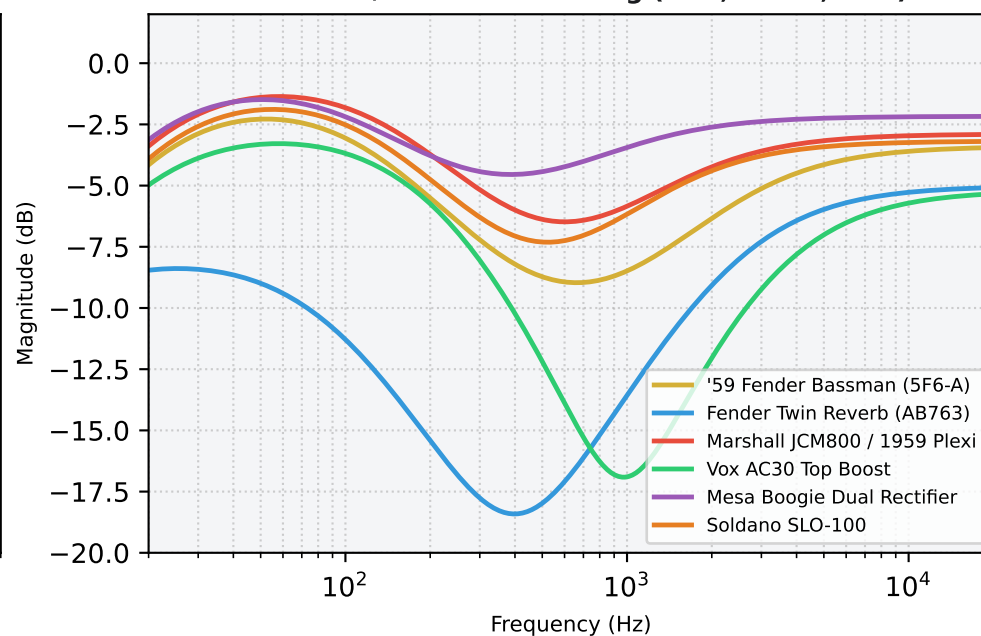
Phase Trajectories across Tone Stacks



Heavy Mid-Scoop Response (T=10, M=0, B=10)



Mid Push / Solo Lead Setting (T=5, M=10, B=3)



# CIRCUIT COMPONENT SPECIFICATIONS & ACOUSTIC MAPPING

Mathematical Component Values and Characteristic Notch Frequencies

Ready for C++ Modular Implementation

| Amplifier Circuit         | R1 (Treb) | R2 (Bass) | R3 (Mid) | R4 (Slope) | C1 (Treb) | C2 (Bass) | C3 (Mid) | Notch Freq               | Acoustic Personalities                             |
|---------------------------|-----------|-----------|----------|------------|-----------|-----------|----------|--------------------------|----------------------------------------------------|
| '59 Fender Bassman (5E6)  | 250k      | 1.0M      | 25k      | 56k        | 250 pF    | 20 nF     | 20 nF    | Original 725 Hz          | 725 Hz benchmark; balanced warmth, 80% treble      |
| Fender Twin Reverb (AB76) | 250k      | 250k      | 10k      | 100k       | 250 pF    | 100 nF    | 47 nF    | Classic Blackface 400 Hz | 400 Hz huge low-end authority, deeper scooped mids |
| Marshall JCM800 / 1959    | 220k      | 1.0M      | 22k      | 33k        | 470 pF    | 22 nF     | 22 nF    | British crunch 672 Hz    | 672 Hz forward midrange, higher treble bite        |
| Vox AC30 Top Boost        | 1000k     | 1.0M      | 10k      | 100k       | 47 pF     | 22 nF     | 10 nF    | British chirp 908 Hz     | 908 Hz small treble cap, 100k slope, sparkling     |
| Mesa Boogie Dual Rect     | 250k      | 1.0M      | 50k      | 47k        | 680 pF    | 20 nF     | 20 nF    | Modern metal 620 Hz      | 620 Hz treble cap, 50k mid pot for extreme         |
| Soldano SLO-100           | 250k      | 1.0M      | 25k      | 47k        | 470 pF    | 20 nF     | 20 nF    | Boutique 575 Hz          | 575 Hz gain; legendary singing lead, smooth        |